Q1. Why can’t we just ignore censoring?

Censoring means that a patient was alive at the last follow-up, so we know they lived at least that long, but we do not know their exact death time. For a patient censored at 0.3 years, we know they survived until at least 0.3 years. If we incorrectly treated 0.3 years as their death time, the model would make them appear to be at higher risk than the evidence supports. If we deleted all censored patients, we would retain only patients with observed deaths, disproportionately excluding patients who were alive at their last follow-up. Both shortcuts introduce bias: treating censored times as deaths exaggerates early mortality, while deleting censored patients produces a selected sample and distorts survival estimates.

Q2. Why group cross-validation by centre?

Death rates vary widely across centres, from 11% at ROM to 92% at MUV. In random five-fold cross-validation, patients from KI could appear in both the training and validation folds. If the model learns centre-specific patterns, the validation score may look better because the model has already seen that centre. This may not help on KYW, because all 1,193 test patients come from KYW, which is absent from training. Therefore, random cross-validation would likely produce a higher, more optimistic score than centre-grouped cross-validation. Grouping by centre better predicts leaderboard performance because it reflects the real task: predicting outcomes for patients from an unseen hospital.

Q3. What could go wrong with MONOCYTES?

MONOCYTES are missing for 18.1% of training patients but 74.0% of test patients. If we fill missing test values with the training median of 0.37, the calculation is: $0.74 \times 1{,}193 = 882.82 \approx 883$. Therefore, approximately 883 test patients would receive the imputed value of 0.37. If the model learned that higher monocyte levels indicate higher risk, all these patients would receive the same contribution from this variable, so monocytes could not distinguish their risks from one another. Furthermore, the training median is 0.37 while the test median is 0.20, suggesting that the training median may not represent the test centre well. To reduce the problem, we could add a missingness indicator to tell the model which values were imputed, or remove MONOCYTES if it proves unreliable under this large difference in missingness.